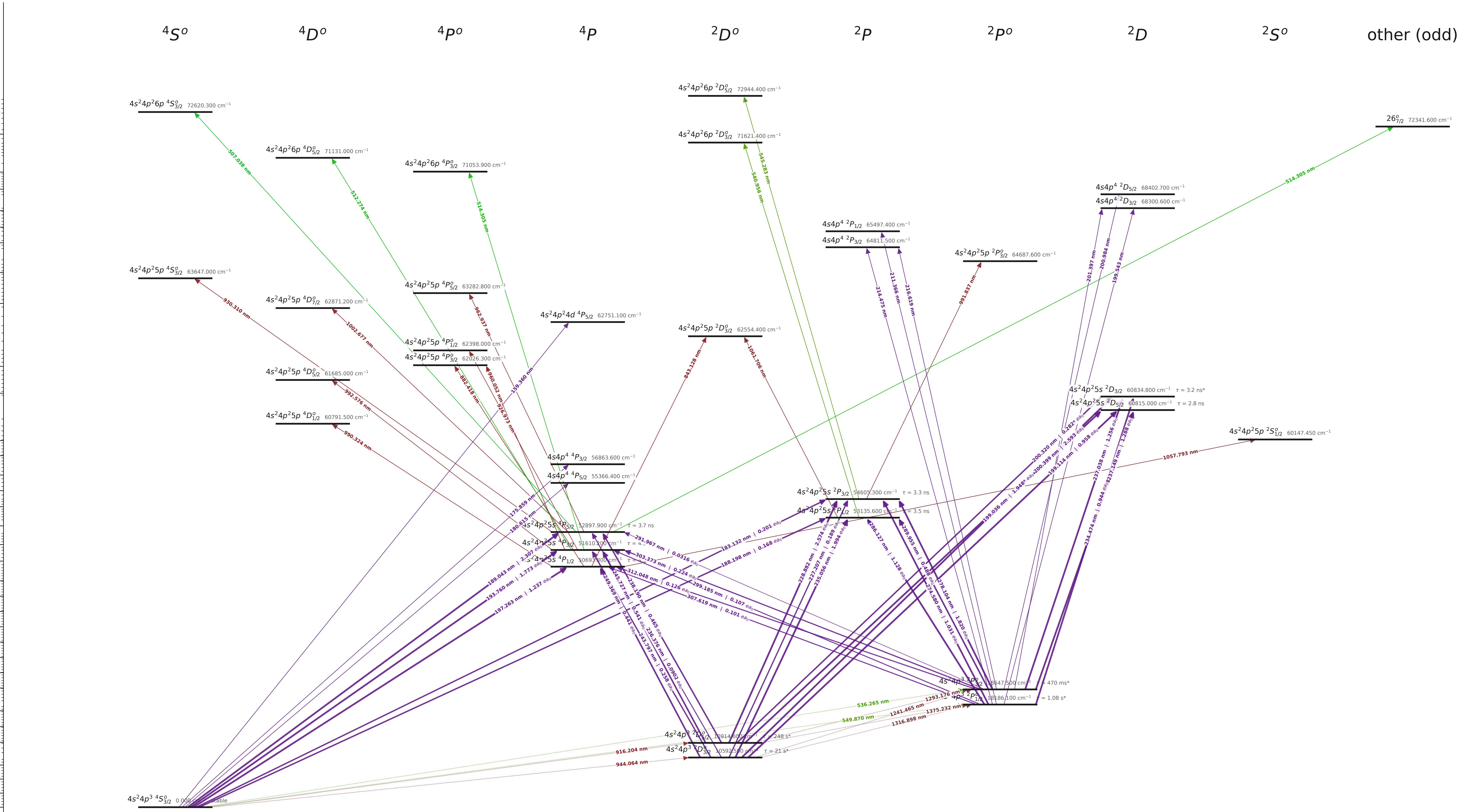


As⁺ ionisation limit 78950.00 cm⁻¹ = 9.78855 eV (126.662 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| \mathbf{er} \| J' \rangle|$ in $e a_0$
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J \mathbf{e} \mathbf{J}\rangle $ (ea)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$4s^2 4p^3 \ ^4S_{3/2} - 4s^2 4p^3 5s \ ^4P_{3/2}$	193.7601	-	1547.2349	1.773	$2.190e+08$	37.01	94	resonance line
$4s^2 4p^3 \ ^4S_{3/2} - 4s^2 4p^3 5s \ ^4P_{1/2}$	197.2628	-	1519.7619	1.237	$2.020e+08$	35.37	94	resonance line
$4s^2 4p^3 \ ^4S_{3/2} - 4s^2 4p^3 \ ^2P_{3/2}^o$	536.2649	536.1158	559.0380	-	$1.600e+00$	$3.386e-07$	75.2	forbidden line within the ground configuration
$4s^2 4p^3 \ ^4S_{3/2} - 4s^2 4p^3 \ ^2P_{1/2}^o$	549.8705	549.7178	545.2056	-	$6.910e-01$	$1.474e-07$	74.6	forbidden line within the ground configuration
$4s^2 4p^3 \ ^4S_{3/2} - 4s^2 4p^3 \ ^2D_{5/2}^o$	916.2040	915.9526	327.2115	-	$5.300e-03$	$6.418e-10$	100	forbidden line within the ground configuration
$4s^2 4p^3 \ ^4S_{3/2} - 4s^2 4p^3 \ ^2D_{3/2}^o$	944.0642	943.8053	317.5552	-	$7.490e-02$	$7.579e-09$	100	forbidden line within the ground configuration
$4s^2 4p^3 \ ^2D_{3/2}^o - 4s^2 4p^3 \ ^2P_{3/2}^o$	1241.4649	1241.1253	241.4828	-	$6.720e-01$	$3.386e-07$	31.6	forbidden line within the ground configuration
$4s^2 4p^3 \ ^2D_{5/2}^o - 4s^2 4p^3 \ ^2P_{3/2}^o$	1293.1759	1292.8223	231.8265	-	$4.400e-01$	$3.386e-07$	20.7	forbidden line within the ground configuration
$4s^2 4p^3 \ ^2D_{5/2}^o - 4s^2 4p^3 \ ^2P_{1/2}^o$	1316.8984	1316.5384	227.6504	-	$4.200e-01$	$1.474e-07$	45.4	forbidden line within the ground configuration
$4s^2 4p^3 \ ^2D_{3/2}^o - 4s^2 4p^3 \ ^2P_{1/2}^o$	1375.2321	1374.8561	217.9941	-	$5.900e-02$	$1.474e-07$	6.37	forbidden line within the ground configuration

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)			
$4s^2 4p^3 \ ^4S_{3/2}$	-66.2040(10)	-0.535(3)	F=0: +247.60	F=1: +181.93	F=2: +50.05	F=3: -149.09
$4s^2 4p^3 \ ^2D_{3/2}$	480(9)	-	F=0: -1800.00	F=1: -1320.00	F=2: -360.00	F=3: +1080.00
$4s^2 4p^3 \ ^2D_{5/2}$	725(26)	-	F=1: -3806.25	F=2: -2356.25	F=3: -181.25	F=4: +2718.75
$4s^2 4p^3 \ ^2P_{1/2}^o$	1909(14)	-	F=1: -2386.25	F=2: +1431.75		
$4s^2 4p^3 \ ^2P_{3/2}^o$	359(59)	-	F=0: -1346.25	F=1: -987.25	F=2: -269.25	F=3: +807.75
$4s^2 4p^2 5s \ ^4P_{1/2}$	609(15)	-	F=1: -760.75	F=2: +456.45		
$4s^2 4p^2 5s \ ^4P_{3/2}$	303(18)	-	F=0: -1135.50	F=1: -832.70	F=2: -227.10	F=3: +681.30
$4s^2 4p^2 5s \ ^4P_{5/2}$	734(12)	-	F=1: -3856.12	F=2: -2387.12	F=3: -183.62	F=4: +2754.38
$4s^2 4p^2 5s \ ^2P_{1/2}$	102(27)	-	F=1: -127.38	F=2: +76.42		
$4s^2 4p^2 5s \ ^2P_{3/2}$	597(15)	-	F=0: -2237.25	F=1: -1640.65	F=2: -443.45	F=3: +1342.35
$4s^2 4p^2 5s \ ^2D_{5/2}$	833(18)	-	F=1: -4375.35	F=2: -2708.55	F=3: -208.35	F=4: +3125.25
$4s^2 4p^2 5s \ ^2D_{3/2}$	720(90)	-	F=0: -2698.12	F=1: -1978.62	F=2: -539.62	F=3: +1618.88

Measured A, B constants; sources in the data file.